

Dot Product	Cross Product of Conformable Matrices
Matrix-Vector Multiplication	Matrix-Matrix Multiplication
Range/Column Space	Nullspace
Rank	Theorem 1.2
Nonsingular/Invertible	A Matrix Inverse Times a Vector

$\mathbf{A}^\top \mathbf{B}$	$\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$
For the matrix-matrix product $B = AC$, each column of B is a linear combination of the columns of A. Let A be $l \times m$, C be $m \times n$, then B is $l \times n$ with entries defined by $b_{ij} = \sum_{k=1}^m a_{ik} c_{kj}$ $\mathbf{b}_j = \mathbf{A} \mathbf{c}_j = \sum_{k=1}^m c_{kj} \mathbf{a}_k$	Let x be an n-dimensional column vector and let A be an $m \times n$ matrix. Then the matrix-vector product $b = Ax$ is the m-dimensional column vector defined as follows: $b_i = \sum_{j=1}^n a_{ij} x_j, \quad i = 1, \dots, m$ $\mathbf{b} = \mathbf{A} \mathbf{x} = \sum_{j=1}^n x_j \mathbf{a}_j$
The nullspace of $\mathbf{A} \in \mathbb{C}^{m \times n}$, written $\text{null}(\mathbf{A})$, is the set of vectors $\mathbf{x}$ that satisfy $\mathbf{A} \mathbf{x} = \mathbf{0}$, where $\mathbf{0}$ is the 0-vector in $\mathbb{C}^m$	• The range of a matrix $\mathbf{A}$, written $\text{range}(\mathbf{A})$, is the set of vectors that can be expressed as $\mathbf{A} \mathbf{x}$ for some $\mathbf{x}$. • The space spanned by the columns of $\mathbf{A}$
A matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$ with $m \geq n$ has full rank if and only if it maps no two distinct vectors to the same vector	Let $\mathbf{A}$ be an $m \times n$ matrix such that $m \geq n$. Then $\mathbf{A}$ has n linearly independent columns with rank n
• $\mathbf{A}^{-1} \mathbf{b}$ is the vector of coefficients of the expansion of $\mathbf{b}$ in the basis of columns of $\mathbf{A}$ • NOTE: Multiplication by $\mathbf{A}^{-1}$ is a change of basis operation	A matrix is invertible if it is a square matrix of full rank

Complex Conjugate	Hermitian Matrix
Inner Product	Euclidean Length (Euclidean Norm)
Cosine of Angle	Orthogonal Vectors
Theorem 2.1	Unitary Matrix
Multiplication by a Unitary Matrix	Norm

A matrix $\mathbf{A}$ is hermitian if $\mathbf{A} = \mathbf{A}^*$	The complex conjugate of a scalar z, written $\bar{z}$ or z^*, is obtained by negating the imaginary part.
$\ \mathbf{x}\ = \sqrt{\mathbf{x}^* \mathbf{x}} = \left(\sum_{i=1}^m x_i ^2 \right)^{1/2}$	The inner product of two column vectors, $\mathbf{x}, \mathbf{y} \in \mathbb{C}^m$ is the product of the adjoint of $\mathbf{x}$ by $\mathbf{y}$: $\mathbf{x}^* \mathbf{y} = \sum_{i=1}^m \bar{x}_i y_i$
A pair of vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal if $\mathbf{x}^* \mathbf{y} = 0$	$\cos \alpha = \frac{\mathbf{x}^* \mathbf{y}}{\ \mathbf{x}\ \ \mathbf{y}\ }$
A square matrix $\mathbf{Q} \in \mathbb{C}^{m \times m}$ is unitary if $\mathbf{Q}^* = \mathbf{Q}^{-1} \iff \mathbf{Q}^* \mathbf{Q} = \mathbf{I}$	The vectors in an orthogonal set S are linearly independent
A norm is a function $\ \cdot\ : \mathbb{C}^m \mapsto \mathbb{R}$ that assigns a real-valued length to each vector. It satisfies the following three properties for all vectors $\mathbf{x}$ and $\mathbf{y}$ and for all scalars $\alpha \in \mathbb{C}$ 1. $\ \mathbf{x}\ \geq 0$ 2. $\ \mathbf{x} + \mathbf{y}\ \leq \ \mathbf{x}\ + \ \mathbf{y}\$ 3. $\ \alpha \mathbf{x}\ = \alpha \ \mathbf{x}\$	• $\mathbf{Q}^* \mathbf{b}$ is the vector of coefficients of the expansion of $\mathbf{b}$ in the basis of columns of $\mathbf{Q}$ • The geometric structure is preserved in this case meaning – $(\mathbf{Q}\mathbf{x})^* (\mathbf{Q}\mathbf{y}) = \mathbf{x}^* \mathbf{y}$ – $\ \mathbf{Q}\mathbf{x}\ = \ \mathbf{x}\$

Normalizing	Most Important Vector Norms
Weighted p -norms	Hölder's Inequality
Cauchy-Schwarz Inequality	Frobenius Norm
Theorem 3.1	The SVD is Motivated by the Following Geometric Fact
NOTE:	Singular Values of A

• $\ \mathbf{x}\ _1 = \sum_{i=1}^m x_i$. Diamond-shaped • $\ \mathbf{x}\ _2 = (\sum_{i=1}^m x_i ^2)^{1/2}$. Circle-shaped • $\ \mathbf{x}\ _\infty = \max_{1 \leq i \leq m} x_i$. Square-shaped • $\ \mathbf{x}\ _p = (\sum_{i=1}^m x_i ^p)^{1/p}$, $1 \leq p < \infty$. Rounded-square-shaped	$\mathbf{v} \mapsto \frac{\mathbf{v}}{\ \mathbf{v}\ }$
Let p and q satisfy $1/p + 1/q = 1$, with $1 \leq p, q \leq \infty$. Then for any vectors $\mathbf{x}$ and $\mathbf{y}$, $ \mathbf{x}^* \mathbf{y} \leq \ \mathbf{x}\ _p \ \mathbf{y}\ _q$	• Each of the coordinates of a vector space is given its own weight • Weighted 2-norm example: $\ \mathbf{x}\ _W = \ W\mathbf{x}\ = \left(\sum_{i=1}^m w_i x_i ^2 \right)^{1/2}$ Horizontal-oval-shaped
$\ \mathbf{A}\ _F = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij} ^2 \right)^{1/2}$	Special case of Hölder's inequality where $p = q = 2$ $ \mathbf{x}^* \mathbf{y} \leq \ \mathbf{x}\ _2 \ \mathbf{y}\ _2$
The image of the unit sphere under any $m \times n$ matrix is a hyperellipse	For any $\mathbf{A} \in \mathbb{C}^{m \times n}$ and unitary $\mathbf{Q} \in \mathbb{C}^{m \times m}$, we have • $\ \mathbf{QA}\ _2 = \ \mathbf{A}\ _2$ • $\ \mathbf{QA}\ _F = \ \mathbf{A}\ _F$
• The lengths of the n principal semiaxes of $\mathbf{AS}$, written $\sigma_1, \sigma_2, \dots, \sigma_n$. • It is conventional to assume that the singular values are numbered in descending order, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$	Look at page 26 of T&B to see the geometry of the SVD

Left Singular Vectors of A	Right Singular Vectors of A
Reduced SVD	Full SVD
Theorem 4.1	Eigenvalue Decomposition
Theorem 5.1	Theorem 5.2
Theorem 5.3	Theorem 5.4

The unit vectors $\{v_1, v_2, \dots, v_n\} \in S$ that are the preimages of the principal semiaxes of $\mathbf{A}S$, numbered so that $\mathbf{A}v_j = \sigma_j u_j$	• The unit vectors $\{u_1, u_2, \dots, u_n\}$ oriented in the directions of the principal semiaxes of $\mathbf{A}S$, numbered to correspond with the singular values. • NOTE: The vector $\sigma_i u_i$ is the ith largest principal semiaxis of $\mathbf{A}S$
Let $\mathbf{A}$ be $m \times n$ with full rank and $m \geq n$ $\mathbf{A} = \mathbf{U}\mathbf{\hat{\Sigma}}\mathbf{V}^*$ • $\mathbf{U}$ is $m \times m$ and unitary • $\mathbf{\hat{\Sigma}}$ is $m \times n$ and diagonal with positive real entries • $\mathbf{V}$ is $n \times n$ and unitary	Let $\mathbf{A}$ be $m \times n$ with full rank and $m \geq n$ $\mathbf{A} = \hat{\mathbf{U}}\hat{\mathbf{\Sigma}}\mathbf{V}^*$ • $\hat{\mathbf{U}}$ is $m \times n$ with orthonormal columns • $\hat{\mathbf{\Sigma}}$ is $n \times n$ and diagonal with positive real entries • $\mathbf{V}$ is $n \times n$ with orthonormal columns AND is unitary
If the columns of a matrix $\mathbf{X} \in \mathbb{C}^{m \times m}$ contain linearly independent eigenvectors of $\mathbf{A} \in \mathbb{C}^{m \times m}$, then the eigenvalue decomposition of $\mathbf{A}$ is $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$ where $\mathbf{\Lambda}$ is an $m \times m$ diagonal matrix whose entries are the eigenvalues of $\mathbf{A}$	Every matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$ has a SVD. Furthermore, the singular values $\{\sigma_j\}$ are uniquely determined.
(MATRIX PROPERTY VIA THE SVD) • $\text{range}(\mathbf{A}) = \langle u_1, \dots, u_r \rangle$ • $\text{null}(\mathbf{A}) = \langle v_{r+1}, \dots, v_n \rangle$	(MATRIX PROPERTY VIA THE SVD) The rank of $\mathbf{A}$ is r, the number of nonzero singular values
(MATRIX PROPERTY VIA THE SVD) The nonzero singular values of $\mathbf{A}$ are the square roots of the nonzero eigenvalues of $\mathbf{A}^*\mathbf{A}$ or $\mathbf{A}\mathbf{A}^*$ (These matrices have the same nonzero eigenvalues).	(MATRIX PROPERTY VIA THE SVD) • $\ \mathbf{A}\ _2 = \sigma_1$ • $\ \mathbf{A}\ _F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}$

Theorem 5.5	Theorem 5.6
Theorem 5.7	Projector
NOTE:	Complementary Projector
Orthogonal Projector	Theorem 6.1
Projection onto vector a	Projection of a onto b

(MATRIX PROPERTY VIA THE SVD) For $\mathbf{A} \in \mathbb{C}^{m \times m}$, $ \det(\mathbf{A}) = \prod_{i=1}^m \sigma_i$	(MATRIX PROPERTY VIA THE SVD) If $\mathbf{A}^* = \mathbf{A}$, then the singular values of $\mathbf{A}$ are the absolute values of the eigenvalues of $\mathbf{A}$
A square matrix $\mathbf{P}$ that satisfies $\mathbf{P}^2 = \mathbf{P}$ ie it is idempotent	(MATRIX PROPERTY VIA THE SVD) $\mathbf{A}$ is the sum of r rank-one matrices: $\mathbf{A} = \sum_{j=1}^r \sigma_j u_j v_j^*$
If $\mathbf{P}$ is a projector, then $\mathbf{I} - \mathbf{P}$ is the complementary projector to $\mathbf{P}$ • $\text{range}(\mathbf{I} - \mathbf{P}) = \text{null}(\mathbf{P})$ • $\text{range}(\mathbf{P}) = \text{null}(\mathbf{I} - \mathbf{P})$	If $\mathbf{v} \in \text{range}(\mathbf{P})$, then $\mathbf{P}\mathbf{v} = \mathbf{v}$
A projector $\mathbf{P}$ is orthogonal if and only if $\mathbf{P} = \mathbf{P}^*$	A projector that projects onto a subspace S_1 along a space S_2, where S_1 and S_2 are orthogonal. (Warning: orthogonal projectors are not orthogonal matrices)
$\text{proj}_b \mathbf{a} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{b}^\top \mathbf{b}} \mathbf{b}$	$\mathbf{P}_a = \frac{\mathbf{a} \mathbf{a}^*}{\mathbf{a}^* \mathbf{a}}$

Orthogonal Projector onto $\text{range}(A)$	Reduced QR Factorization
Full QR Factorization	Gram-Schmidt Orthogonalization
Theorem 7.1	Theorem 7.2
Solution of $Ax = b$ by QR Factorization	Gram-Schmidt Projections
Modified Gram-Schmidt Algorithm	Triangularization

Let $\mathbf{A}$ be $m \times n$ with full rank and $m \geq n$ $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$ • $\hat{\mathbf{Q}}$ is $m \times n$ with orthonormal columns • $\hat{\mathbf{R}}$ is $n \times n$ and upper-triangular	$\mathbf{P} = \mathbf{A}(\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*$
$\mathbf{v}_j = \mathbf{a}_j - (\mathbf{q}_1^* \mathbf{a}_j) \mathbf{q}_1 - (\mathbf{q}_2^* \mathbf{a}_j) \mathbf{q}_2 - \cdots - (\mathbf{q}_{j-1}^* \mathbf{a}_j) \mathbf{q}_{j-1}$ Then normalize: $\mathbf{q}_j = \frac{\mathbf{v}_j}{\ \mathbf{v}_j\ }$ Then $r_{ij} = \mathbf{q}_i^* \mathbf{a}_j$, $r_{jj} = \left\ \mathbf{a}_j - \sum_{i=1}^{j-1} r_{ij} \mathbf{q}_i \right\ _2$	Let $\mathbf{A}$ be $m \times n$ with full rank and $m \geq n$ $\mathbf{A} = \mathbf{Q}\mathbf{R}$ • $\mathbf{Q}$ is $m \times m$ and its columns $\mathbf{q}_j$ are orthogonal to $\text{range}(\mathbf{A})$ • $\hat{\mathbf{R}}$ is $m \times n$ and upper-triangular
Each $\mathbf{A} \in \mathbb{C}^{m \times n}$ ($m \geq n$) of full rank has a unique reduced QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$ with $r_{jj} > 0$	Every $\mathbf{A} \in \mathbb{C}^{m \times n}$ ($m \geq n$) has a full QR factorization, hence also a reduced QR factorization
Let $\mathbf{A} \in \mathbb{C}^{m \times n}$, $m \geq n$ be a matrix of full rank with columns $\{\mathbf{a}_j\}$ $\mathbf{q}_1 = \frac{\mathbf{P}_1 \mathbf{a}_1}{\ \mathbf{P}_1 \mathbf{a}_1\ }, \dots, \mathbf{q}_n = \frac{\mathbf{P}_n \mathbf{a}_n}{\ \mathbf{P}_n \mathbf{a}_n\ }$ Where $\mathbf{P}_j$ is the orthogonal projector that projects $\mathbb{C}^m$ orthogonally onto the space orthogonal to $\langle \mathbf{q}_1, \dots, \mathbf{q}_{j-1} \rangle$.	1. Compute a QR factorization $\mathbf{A} = \mathbf{Q}\mathbf{R}$ • $\mathbf{A}\mathbf{x} = \mathbf{b} \iff \mathbf{Q}\mathbf{R}\mathbf{x} = \mathbf{b} \iff \mathbf{R}\mathbf{x} = \mathbf{Q}^* \mathbf{b}$ 2. Compute $\mathbf{y} = \mathbf{Q}^* \mathbf{b}$ • $\mathbf{R}\mathbf{x} = \mathbf{Q}^* \mathbf{b} \iff \mathbf{R}\mathbf{x} = \mathbf{y}$ 3. Solve $\mathbf{R}\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$
The matrix $\mathbf{Q}_k$ is chosen to introduce zeros below the diagonal in the kth column while preserving all the zeros previously introduced.	$\mathbf{v}_j = \mathbf{P}_{\perp \mathbf{q}_{j-1}} \cdots \mathbf{P}_{\perp \mathbf{q}_2} \mathbf{P}_{\perp \mathbf{q}_1} \mathbf{a}_j$

Householder Reflector	Householder Triangularization
Overdetermined System	Pseudoinverse
Full-Rank Least Squares Problems can be found with one of the following two vectors	Least Squares by Normal Equations
Least Squares by QR Factorization	Least Squares by SVD
Forward Substitution	Back Substitution

1. Take the first column of $\mathbf{A}$ and construct a Householder reflector $\mathbf{Q}_1$ zeros out every entry below the first diagonal element 2. Left multiply $\mathbf{Q}_1$ to $\mathbf{A}$ 3. Do the same thing for columns $2 : n$ of $\mathbf{A}$ 4. Now we have the matrix $\mathbf{R} = \mathbf{Q}_n \cdots \mathbf{Q}_2 \mathbf{Q}_1 \mathbf{A}$ 5. Let $\mathbf{Q} = \mathbf{Q}_n \cdots \mathbf{Q}_2 \mathbf{Q}_1$. Then $\mathbf{Q}$ is symmetric since it is unitary. Thus we get $\mathbf{A} = \mathbf{Q}\mathbf{R}$	$\mathbf{Q}_k = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{F} \end{bmatrix}$ Where $\mathbf{I}$ is the $(k-1) \times (k-1)$ identity matrix and $\mathbf{F}$ is a $(m-k+1) \times (m-k+1)$ unitary matrix.
$\mathbf{A}^+ = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^* \in \mathbb{C}^{n,m}$	• A rectangular system of equations with $m > n$ • More equations than unknowns
1. Form the matrix $\mathbf{A}^* \mathbf{A}$ and the vector $\mathbf{A}^* \mathbf{b}$ 2. Compute the Cholesky factorization $\mathbf{A}^* \mathbf{A} = \mathbf{R}^* \mathbf{R}$ 3. Let $\mathbf{w} = \mathbf{R}\mathbf{x}$. Use forward substitution to solve lower-triangular system $\mathbf{R}^* \mathbf{w} = \mathbf{A}^* \mathbf{b}$ for $\mathbf{w}$. 4. Use back substitution to solve upper-triangular system $\mathbf{R}\mathbf{x} = \mathbf{w}$ for $\mathbf{x}$	• $\mathbf{x} = \mathbf{A}^+ \mathbf{b}$ • $\mathbf{y} = \mathbf{P}\mathbf{b}$
1. Note that the orthogonal projector $\mathbf{P}$ in $\mathbf{y} = \mathbf{P}\mathbf{b}$ has factorization $\mathbf{P} = \hat{\mathbf{U}}\hat{\mathbf{U}}^*$. 2. Also note that $\mathbf{y} \in \text{range}(\mathbf{A})$, so $\mathbf{A}\mathbf{x} = \mathbf{y}$ has an exact solution. Furthermore, $\mathbf{A}$ has SVD factorization $\mathbf{A} = \hat{\mathbf{U}}\hat{\Sigma}\mathbf{V}^*$ 3. $\mathbf{y} = \mathbf{P}\mathbf{b} \iff \hat{\mathbf{U}}\hat{\Sigma}\mathbf{V}^* \mathbf{x} = \hat{\mathbf{U}}\hat{\mathbf{U}}^* \mathbf{b} \iff \hat{\Sigma}\mathbf{V}^* \mathbf{x} = \hat{\mathbf{U}}^* \mathbf{b}$ 4. Let $\mathbf{w} = \mathbf{V}^* \mathbf{x}$. Solve the diagonal system $\hat{\Sigma}\mathbf{w} = \hat{\mathbf{U}}^* \mathbf{b}$ for $\mathbf{w}$. Then set $\mathbf{x} = \mathbf{V}\mathbf{w}$	1. Note that the orthogonal projector $\mathbf{P}$ in $\mathbf{y} = \mathbf{P}\mathbf{b}$ has factorization $\mathbf{P} = \hat{\mathbf{Q}}\hat{\mathbf{Q}}^*$. 2. Also note that $\mathbf{y} \in \text{range}(\mathbf{A})$, so $\mathbf{A}\mathbf{x} = \mathbf{y}$ has an exact solution. Furthermore, $\mathbf{A}$ has QR factorization $\mathbf{A} = \hat{\mathbf{Q}}\hat{\mathbf{R}}$ 3. $\mathbf{y} = \mathbf{P}\mathbf{b} \iff \hat{\mathbf{Q}}\hat{\mathbf{R}}\mathbf{x} = \hat{\mathbf{Q}}\hat{\mathbf{Q}}^* \mathbf{b} \iff \hat{\mathbf{R}}\mathbf{x} = \hat{\mathbf{Q}}^* \mathbf{b}$ 4. Use back substitution to solve the upper triangular system $\hat{\mathbf{R}}\mathbf{x} = \hat{\mathbf{Q}}^* \mathbf{b}$ for $\mathbf{x}$
Let $\mathbf{U}$ be an upper triangular matrix. Solve $\mathbf{U}\mathbf{x} = \mathbf{b}$ (where $\mathbf{U} \in \mathbb{C}^{m \times m}$) for $\mathbf{x}$ using forward substitution: $x_i = \left(b_i - \sum_{j=i+1}^m u_{ij}x_j \right) / u_{ii}$ where $x_m = b_m / u_{mm}$	Let $\mathbf{L}$ be a lower triangular matrix. Solve $\mathbf{L}\mathbf{x} = \mathbf{b}$ for $\mathbf{x}$ using forward substitution: $x_i = \left(b_i - \sum_{t=1}^{i-1} l_{it}x_t \right) / l_{ii}$ where $x_1 = b_1 / l_{11}$

LU Factorization	Helpful Taxonomy of Factorization Algorithms
NOTE:	Gaussian Elimination without Pivoting
Solving Linear Equations ($Ax = b$) by LU Factorization	NOTE:
Positive Definite	NOTE:
Cholesky Factorization	Theorem 23.1

• Gram-Schmidt: $\mathbf{A} = \mathbf{QR}$ by triangular orthogonalization • Householder: $\mathbf{A} = \mathbf{QR}$ by orthogonal triangularization • Gaussian Elimination: $\mathbf{A} = \mathbf{LU}$ by triangularization	• Requires that $\mathbf{A} \in \mathbb{C}^{m \times m}$ • Similar to Householder triangularization in that you left multiply a series of lower triangular matrices $\mathbf{L}_k$ into $\mathbf{A}$ to obtain an upper triangular matrix $\mathbf{U}$ • $\mathbf{L}^{-1} = \mathbf{L}_{m-1} \cdots \mathbf{L}_2 \mathbf{L}_1$ • $\mathbf{L}^{-1} \mathbf{A} = \mathbf{U} \iff \mathbf{A} = \mathbf{LU}$
<pre> m <- nrow(A); n <- ncol(A) L = diag(n) for (j in 1:(n-1)) { for (i in (j+1):m) { L[i,j] <- A[i,j] / A[j,j] A[i,] <- A[i,] - L[i,j] * A[j,] } } </pre>	Gaussian elimination requires $\mathbf{A}$ to be a square matrix
Gaussian Elimination is unstable without pivoting	1. Decompose $\mathbf{A} = \mathbf{LU}$. Now we have $\mathbf{LUx} = \mathbf{b}$ 2. Let $\mathbf{y} = \mathbf{Ux}$. Use forward substitution to solve lower triangular system $\mathbf{Ly} = \mathbf{b}$ for $\mathbf{y}$. 3. Use back substitution to solve upper triangular system $\mathbf{y} = \mathbf{Ux}$ for $\mathbf{x}$
If $\mathbf{A}$ is positive definite and $\mathbf{X}$ is an $m \times n$ matrix of full rank with $m \geq n$ then the matrix $\mathbf{X}^* \mathbf{A} \mathbf{X}$ is also positive definite	A matrix $\mathbf{A}$ is positive definite if $\mathbf{x}^* \mathbf{A} \mathbf{x} > 0 \quad \forall \mathbf{x} \neq \mathbf{0}$
Every positive definite matrix $\mathbf{A} \in \mathbb{C}^{m \times m}$ has a unique Cholesky factorization	$\mathbf{A} = \mathbf{R}^* \mathbf{R}$ • $\mathbf{R}$ is upper-triangular

Cholesky Factorization Algorithm	

	<pre>m <- nrow(A) R <- A for (k in 1:m) { if (k < m) for (j in (k+1):m) { R[j,j:m] <- R[j,j:m] - R[k,j:m] * (Conj(R[k,j])/R[k,k]) } R[k,k:m] <- R[k,k:m] / sqrt(R[k,k]) } R * upper.tri(R, diag = TRUE)</pre>